IDENTIFYING DECISION BOUNDARIES TO EXPLAIN BLACK-BOX MACHINE LEARNING PREDICTIONS

Part 1: Background, Motivation & the Case for Counterfactuals

Adriaan Rowan • PhD Thesis Presentation

In this section

- Why Explain a model
- Interpretable vs Black-box
- Local + Global
- Counterfactuals

Why This Topic: Example for loan application models

Both model owner and individuals require interpretations – but their questions and acceptable answers vary

"Why was my loan application rejected?"

A question of understanding — what drove the decision?

"What do I need to change to get approved?"

A question of action — what must change to flip the outcome?

Why This Is Not Optional

Legal, ethical and commercial drivers

The Law Requires It

South Africa's POPI Act requires automated decision models to provide sufficient information about underlying logic to the individuals affected.

Clients Demand It

Insurance, credit and health decisions affect lives. Customers and regulators increasingly demand clear, actionable explanations — not just a score.

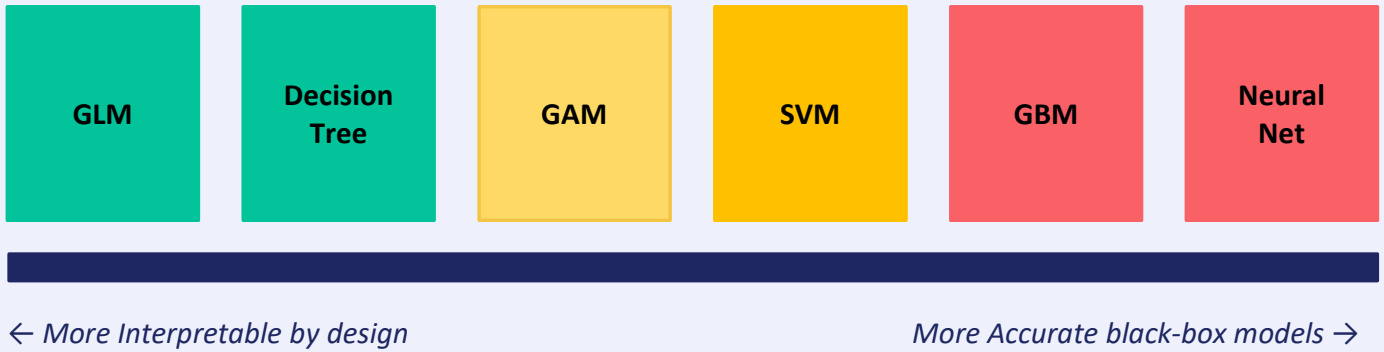
Better Models Result

Interpretability is not the enemy of accuracy.

Understanding a model's behaviour leads to better model performance.

The False Trade-Off

Accuracy vs interpretability — a dilemma that can be resolved



Example: Smoking & Cancer Research

To conclude that smoking causes cancer, you need: (1) an accurate model predicting mortality, (2) a method to isolate each variable's contribution, and (3) a clear definition of 'significance'. Without interpretability, even a perfectly accurate model cannot prove causation.

The Resolution

- Interpretability often BEGETS accuracy, not the reverse
- Explaining a model reveals poor features that should be removed
- Add a model-agnostic explanation layer – to help explain any model
- High-stakes contexts demand both accuracy AND explanation

Four Distinctions That Define the Problem

Getting the terminology right before proposing a solution and giving an explanation

Global interpretation

How the model works overall — feature patterns, boundary shape

vs

Local interpretation

Why THIS specific prediction was made — and what to change

Model-specific method

Works only for one model type (e.g. gradient-based)

vs

Model-agnostic method

Works for any classifier — black box or transparent

Interpretable by design

GLM, decision tree: transparent structure, weaker predictions

vs

Post hoc explanation

Apply explanation layer AFTER fitting any complex model

Understanding WHY

Attribution: which features drove the prediction?

vs

Knowing WHAT TO CHANGE

Recourse: what actionable changes flip the outcome?

The SHAP Problem — A Closer Look

Widely adopted, but it answers a different question than practitioners need

What SHAP claims to answer:

"Why is this prediction different from the average prediction?"

≠

What practitioners need to know:

"What must change for this individual to get a different outcome?"

Wrong reference point

SHAP explains deviation from the global average — but a client doesn't care about the average. They want to know about THEIR situation.

Computationally expensive

Exact Shapley computation is exponential in the number of features. Approximations introduce error and instability in practice.

No actionable path

SHAP tells you what contributed — not what to change. A variable with high SHAP importance may be immutable (e.g. age). No recourse is provided.

Introducing Counterfactuals

The closest possible world where the outcome flips

A counterfactual is the nearest data point to an individual observation — but with the opposite predicted outcome.

ACTUAL APPLICATION

Income:	R 18 000 / month
Monthly debt:	R 12 000 / month
Age:	30 Years old

✗ **DECLINED**

→

COUNTERFACTUAL

Income:	R 20 000 / month ← **increase by R 2 000**
Monthly debt:	R 9 000 / month ← **reduce by R 3 000**
Age:	25 Years old ← **reduce by 5 years**

✓ **APPROVED**

Not Any Change Will Do

Counterfactuals must meet four key constraints to be useful

A

Actionable

Only change features the person can actually influence. Age is immutable. Debt level is actionable. The counterfactual must respect which variables can realistically change.

Do NOT suggest: 'be 5 years younger'

F

Feasible

The proposed values must fall within realistic bounds for each feature. An income of R 500 000/month may be mathematically optimal but practically impossible.

Stay within the training data range per variable

P

Plausible

The combination of feature values must make sense together — not just individually. Higher income & low age is ideas but = implausible. Changing one variable while ignoring correlated variables can produce unrealistic profiles.

Produce counterfactuals that considers the correlation

S

Sparse

Change as few features as possible. A counterfactual that requires changing 10 variables is not actionable. One or two key changes is better than ten small ones

Identify the smallest set of changes that flips the prediction.

Part 2

Understanding Decision Boundaries

What they are, why they differ across models, and why finding them is the key to everything

In this section

- What is a decision boundary?
- The 0.5 threshold
- Activation functions
- Visualising boundaries
- Why complexity is the problem
- Why this unlocks XAI

What Is a Decision Boundary?

The line — or surface — where the model's prediction flips from class 0 to class 1

The purpose of a classification model is to separate class 0 from class 1 data points. The model outputs a score between 0 and 1. The decision boundary is where that score crosses the threshold — **typically 0.5**

Below the threshold → Class 0

If the model scores a point at 0.3, it is assigned to class 0. The point sits on the class 0 side of the decision boundary.

Above the threshold → Class 1

If the model scores a point at 0.7, it is assigned to class 1. The point sits on the class 1 side of the decision boundary.

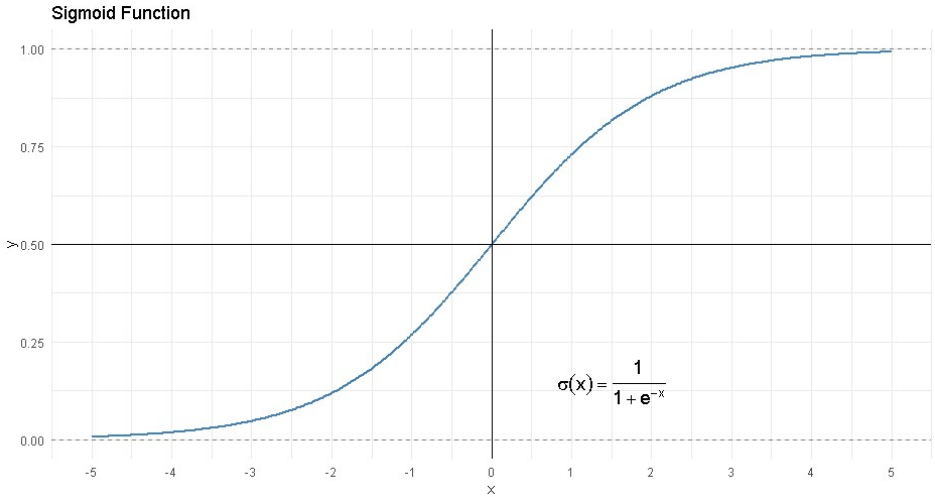
The score is NOT a probability

A score of 0.6 does NOT mean 60% chance of class 1. It is a mathematical optimisation output. Calling it a probability requires calibration — this is frequently misunderstood.

The Shape of the Boundary Depends on the Model

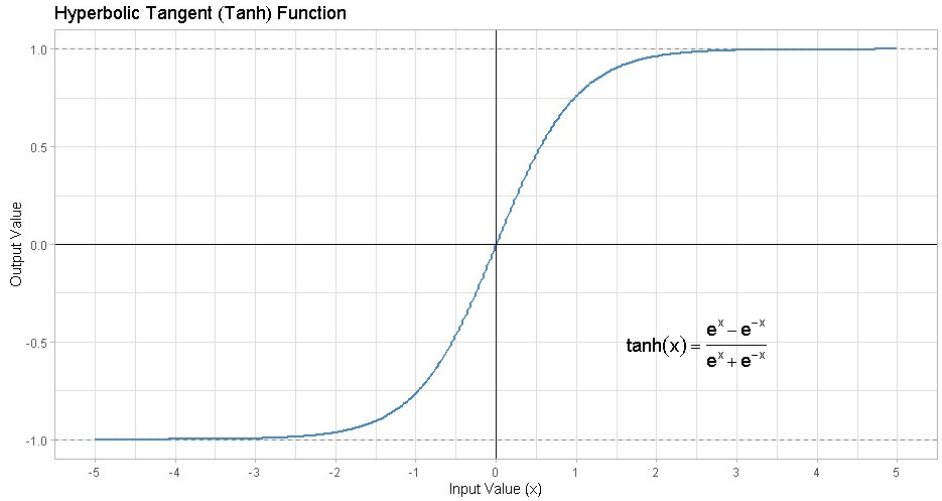
Activation functions determine how the score transitions between 0 and 1 — and therefore where the boundary falls

Sigmoid Function (Logistic Regression)



The boundary is a smooth line or plane where the sigmoid crosses 0.5. For two variables this is a straight line in feature space — a linear decision boundary – with an S-shaped curve

Tanh Function (Neural Networks)

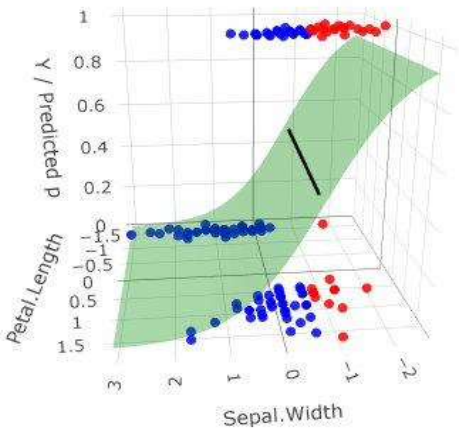


In neural networks, layers of tanh (or other) activations combine to produce highly non-linear decision surfaces. The boundary value is where the tanh crosses 0. The boundary is no longer a simple plane and the gives a steeper curve.

A Linear Boundary — Logistic Regression

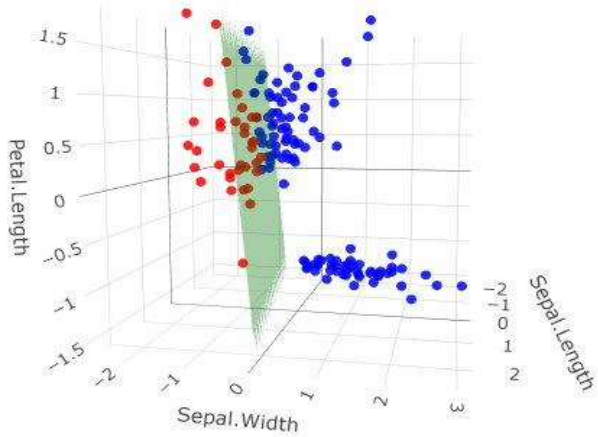
In two variables, the decision boundary is a straight line. In three, it becomes a plane.

2D View — The Decision Boundary as a Line



The green plane shows where the model predicts p = 0.5. The black vertical line marks the decision boundary — where class 0 (blue) meets class 1 (red).

3D View — The Boundary as a Plane in Feature Space

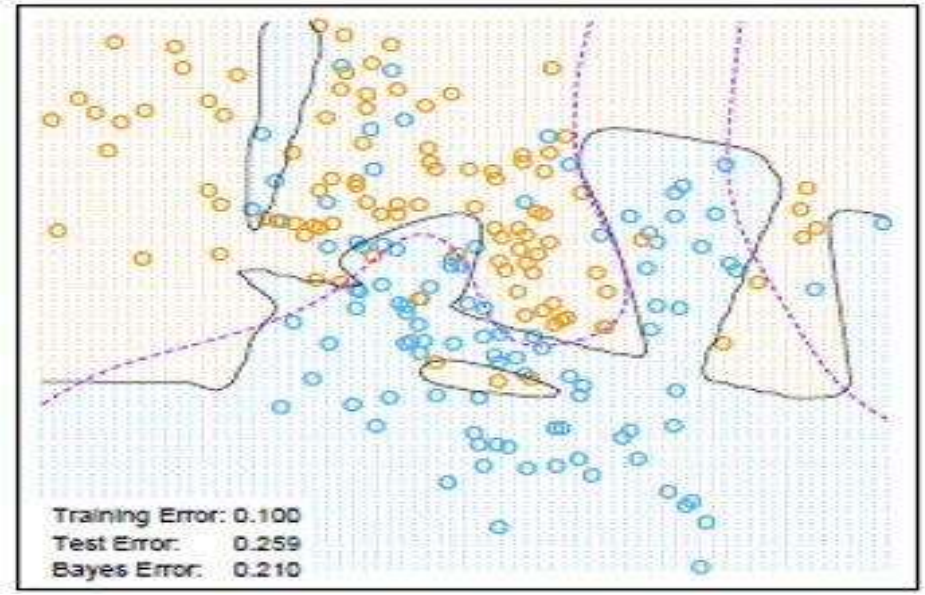


With three input variables the boundary becomes a 2D plane embedded in 3D space. This is still a linear boundary — interpretable and stable.

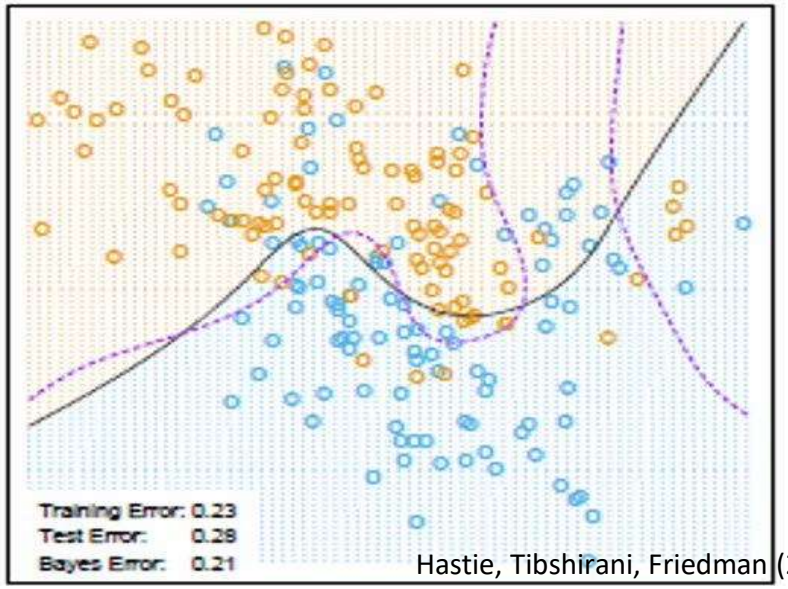
When the Boundary Gets complex it creates Black Boxes

Black-box models can fit highly irregular boundaries — accurate, but impossible to identify or calculate directly

Neural Network — Highly Non-Linear Boundary



GAM / Spline — Smooth but Non-Linear Boundary



Hastie, Tibshirani, Friedman (2009)

In real credit, insurance or health datasets there may be 10, 20 or 50 input variables. The decision boundary lives in that high-dimensional space — and cannot be directly seen or measured. The model only gives a score; it does not provide a route to the boundary, the distance to it, or the direction in which each variable must move to reach it. This is exactly the problem this research solves.

The Problem: The Boundary Is Hidden

The model gives a score — but not a map of where the boundary lies

What the model gives you

A score between 0 and 1

$f(x) = 0.73 \rightarrow$ predicted class 1

No information about distance to boundary

Is this point just over 0.5, or solidly at 0.9?

No variable-level directions to the boundary

Which variable, changed by how much, would flip the prediction?

No map of the boundary shape

Where exactly does the model change its mind — and why?

If we COULD find the boundary...

Find counterfactuals

The nearest point on the boundary IS the nearest counterfactual — the minimal change that flips the prediction.

Explain variable contributions

The path from a data point to the boundary shows exactly which variables need to change — and by how much.

Measure model robustness

Distance to the boundary is a direct measure of prediction stability. Close = fragile. Far = robust.

Cluster and understand structure

Knowing boundary regions reveals how the model divides feature space — giving true algorithmic transparency.

From Decision Boundary to Counterfactual

Once the boundary is known, the counterfactual follows directly

A counterfactual is the closest point to an observation that lies on the other side of the decision boundary — the minimal change that flips the predicted class. (Wachter, Mittelstadt & Russell, 2017)

SHAP

Tells you what contributed to the prediction vs the average

Does NOT tell you what to change. Does not give a counterfactual. Answers a different question entirely.

GLM (Logistic Regression)

Coefficients give direction of each variable's influence

Does not automatically produce a counterfactual. Requires projecting the point onto the boundary — this is not done by default.

Counterfactual (This Method)

Directly answers: what is the minimal change to flip the prediction? Actionable, feasible, sparse 'if-then' statements.

Not attempting to explain the model's internal logic — explains what the individual must do differently.

How to find the boundary…

If we can project high-dimensional data into two dimensions — and preserve the geometric structure — we can approximate the decision boundary visually, find the nearest counterfactual, and recover the required variable changes.

1

Project to 2D

Use biplots to reduce p variables to two dimensions while preserving geometric distances between observations.

→

2

Approximate the boundary

Score a fine grid of 2D points through the model using inverse projection. The contour at score = 0.5 approximates the decision boundary.

→

3

Find counterfactuals

The nearest point on the 2D boundary approximation is the counterfactual. Inverse-project back to original variable values to explain the change.

Part 3: The Biplot Framework →

Part 3

The Biplot Counterfactual Framework

How biplots make decision boundaries visible, counterfactuals findable, and predictions explainable

In this section

Introducing biplots

Visualising the decision boundary

Finding the counterfactual

Global interpretation

Surrogate model

Local interpretation

Limitations

Introducing Biplots

An optimal 2D representation that preserves both distances and variable relationships

What is a Biplot

Normal plot can only show at most three variables at a time – limiting the ability to visualise and investigate the interaction across multiple variables

The 'bi' in biplot refers to the simultaneous display of both observations (points) and variables (calibrated axes) in the same 2D space — allowing values to be read off the axes by projection.

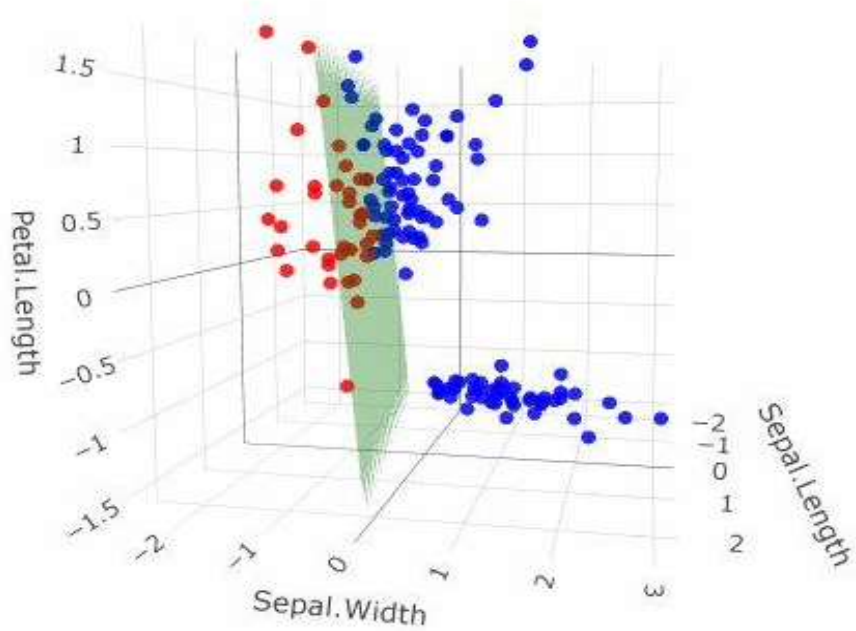
A biplot is an optimal least-squares 2D representation of a multi-dimensional dataset. Euclidean distances in 2D reflect similarity in the original high-dimensional space — points that are similar in all variables appear close together.

$\boldsymbol{X}$: $n \times p = \boldsymbol{U\Sigma V}'$ => Select first two columns of $\boldsymbol{V}$ => $\boldsymbol{V}_2$

Project onto biplot: $\mathrm{Z} = \boldsymbol{XV}_2$; Axes direction from rows in $\boldsymbol{V}_2$

Predict back to n-dimensional data: $\widehat{\boldsymbol{X}} = \boldsymbol{ZV}_2'$

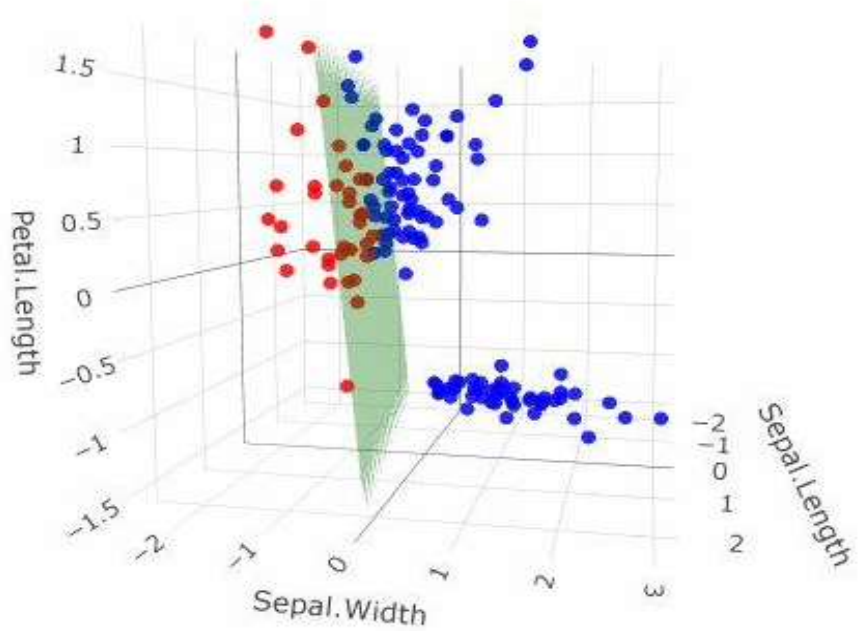
3-Variable Plot with Decision Boundary



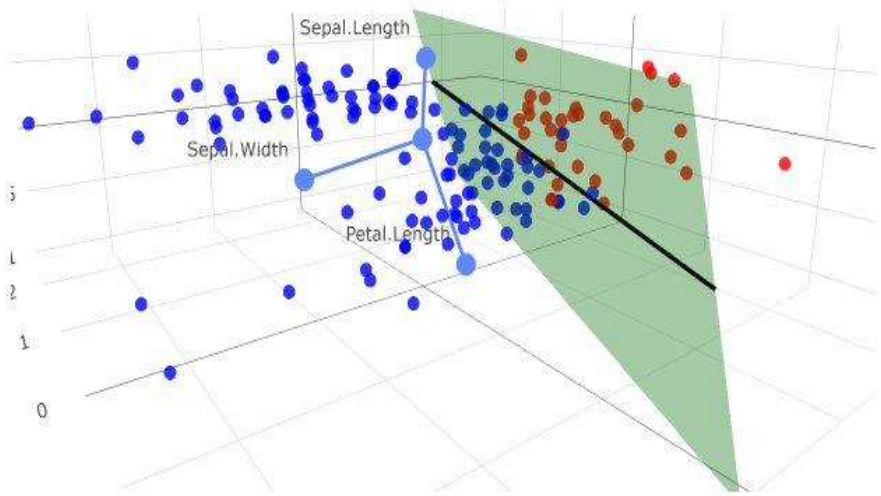
Example of Normal 3D-plot vs 3D-Biplot

Create uncorrelated data points with the variable axes added as separate orthogonal components to the plots

3-Variable Plot with Decision Boundary



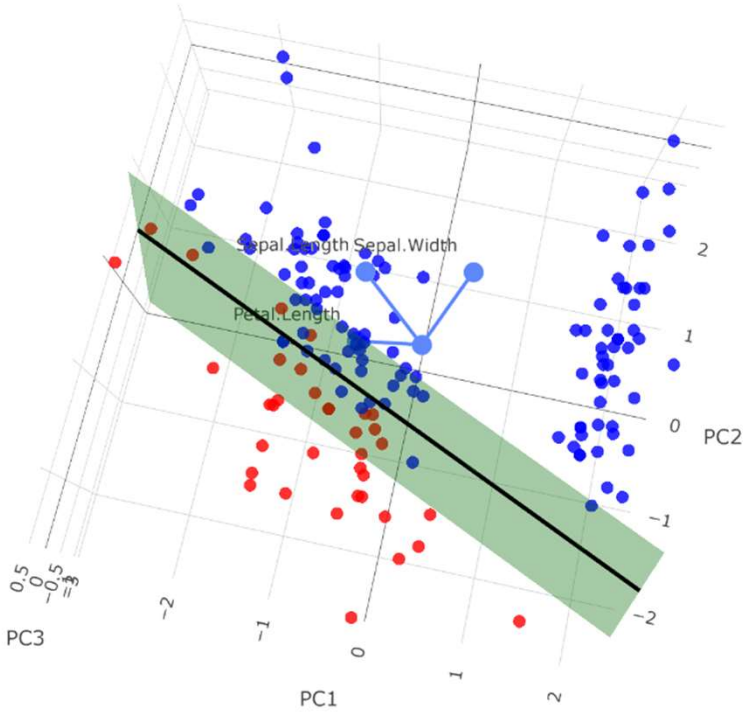
3-Variable data represented by 3D-biplot & Decision boundary



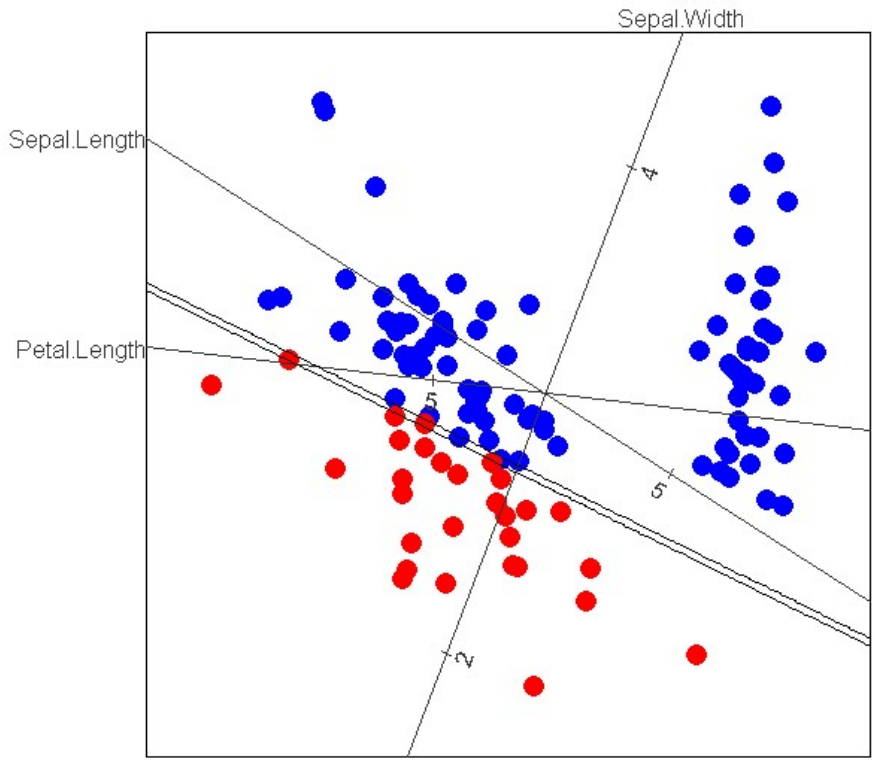
Reducing the biplot to two dimensions

Distance between the data points indicate their similarity, axes direction indicate the correlation between variables

3-Variable data represented by 3D-biplot & Decision boundary

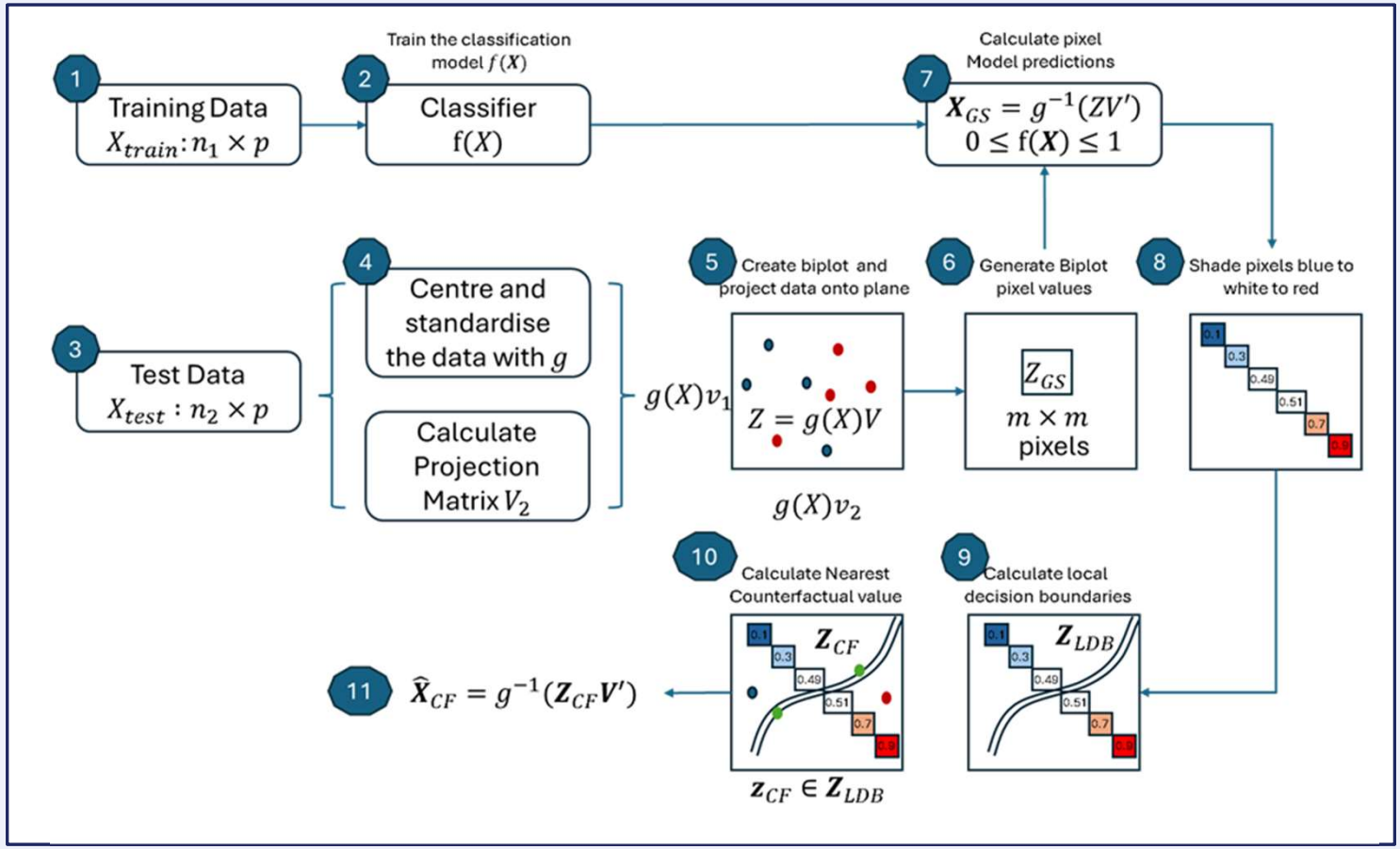


Projecting 3D Biplot Onto a 2D Biplot Plane



Finding the Counterfactual — The Process

An 11-step workflow from training data to identified counterfactual



Steps 1–3

Training data + classifier + test data

Steps 4–5

Centre, standardise, project to 2D (Z = g(X)V)

Steps 6–8

Score m×m grid via inverse projection: $X = g^{-1}(ZV')$

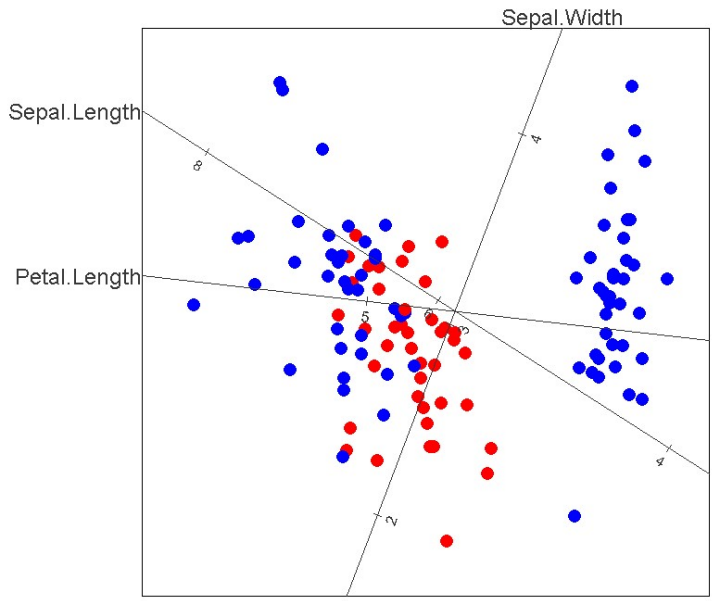
Steps 9–11

Extract contour at 0.5 → local boundary Z_LDB → counterfactual Z_CF

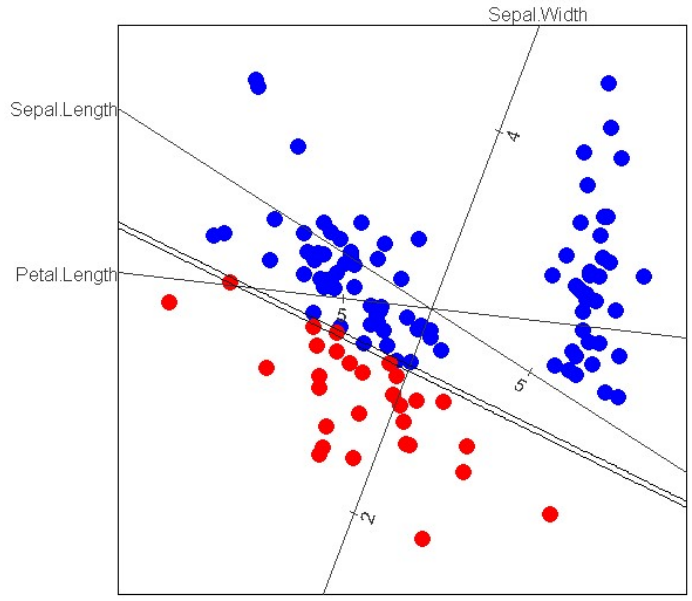
Steps 1- 5: Fitting model and project to 2D

Model fits a classification boundary to the data resulting in data predictions

Biplot of the data coloured by their *original* classification



Biplot of the data coloured by their *predicted* classification



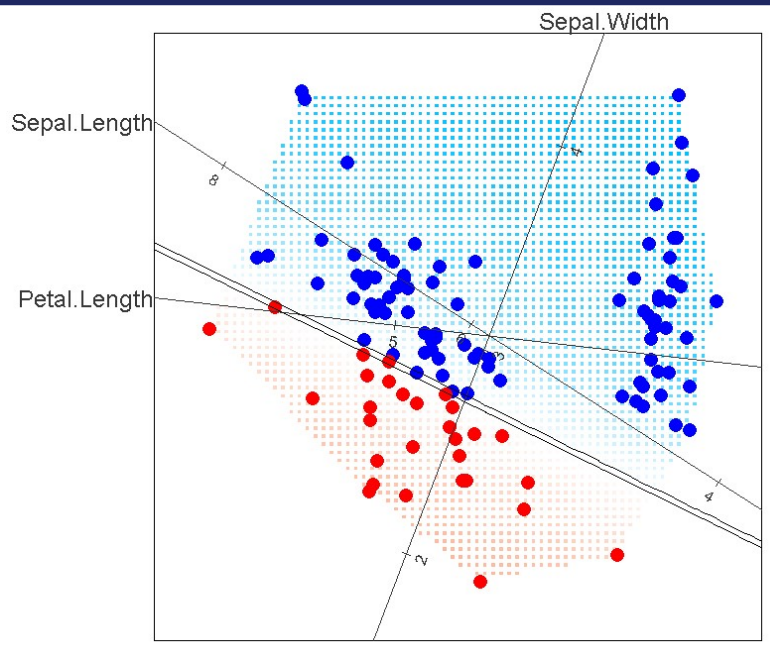
Machine learning model aims to find the optimal separation between the two classes of target variable in the data, subject to the model design constraints.

Data coloured on the biplot to reflect their predicted class. Comparison of the distribution of the original vs predicted classes indicates why a model has different prediction accuracy

Steps 6-8 Finding Local Decision Boundaries

Each observation has its own nearest boundary segment — its local decision boundary

Create a grid of datapoints on the biplot plane

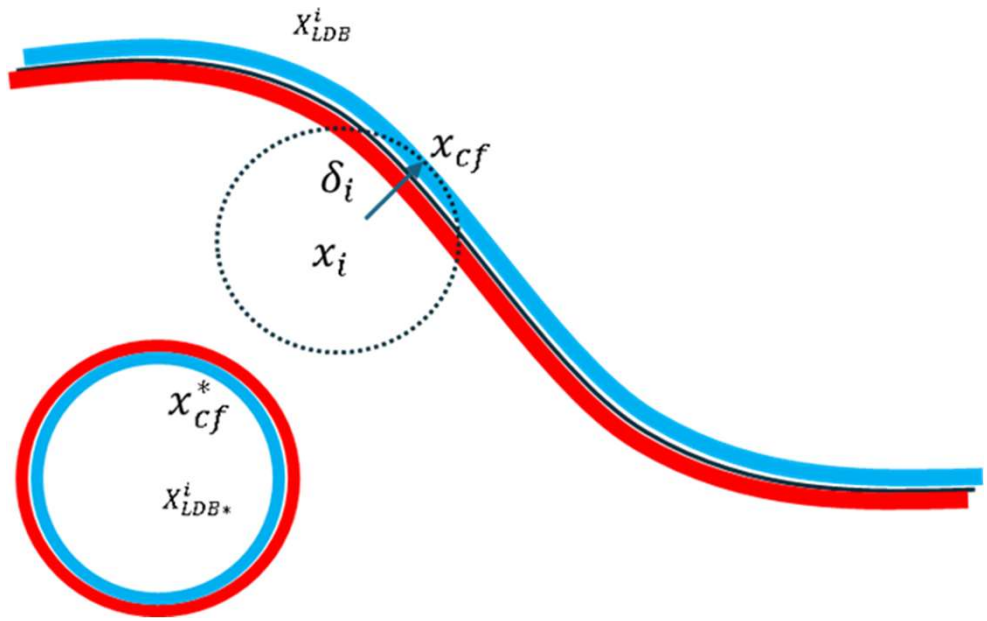


The grid points are predicted back to n-dimensional data as $\widehat{\boldsymbol{X}} = \boldsymbol{Z}\boldsymbol{V}_2'$.

These are then predicted by the model as a score between 0 and 1.

Decision boundary $\boldsymbol{Z}_{LDB}$ represented by contour lines along 0.5 +/- 0.01.

Multiple observations — shared boundaries



The dashed circle shows the search radius of each data point.

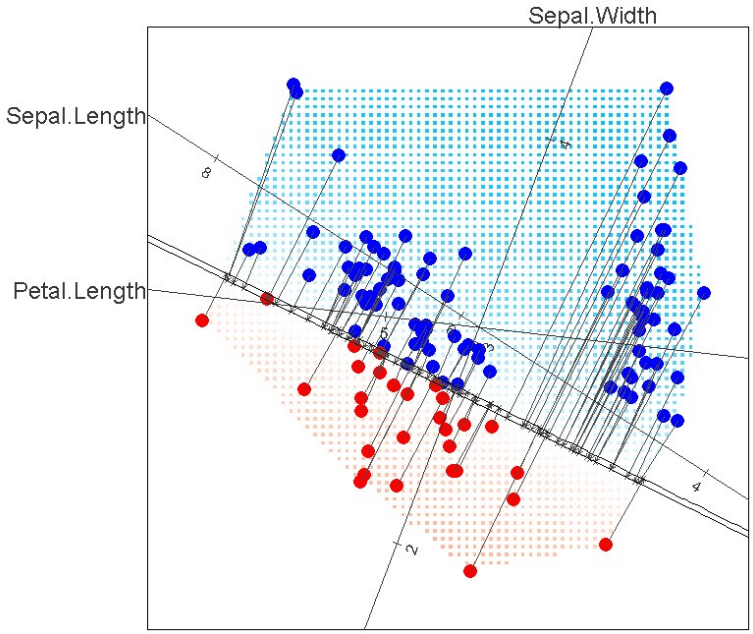
The nearest point on $\boldsymbol{Z}_{LDB}$ represents the counterfactual $\boldsymbol{x}_{cf}$.

Choose the nearest point between different sets of local boundaries.

Steps 9-11 Local Decision Boundaries and Counterfactuals

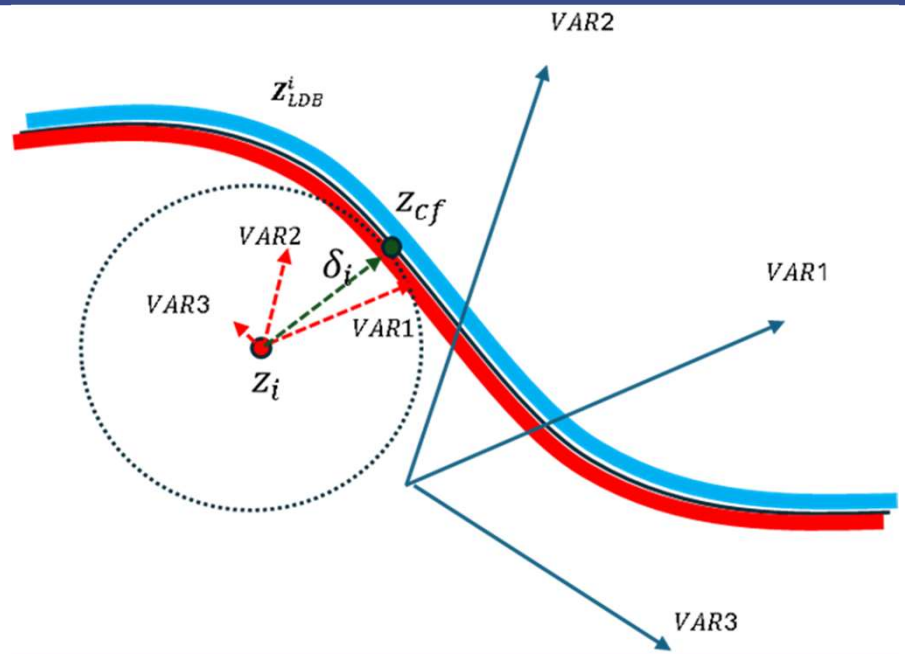
The Local Decision Boundary gives the counterfactual, which can be decomposed into the contribution by each variable

Find the change vector to the counterfactual in 2D-space



Change from the data point to its nearest counterfactual is represented by the vector $\boldsymbol{\delta}_i = (\boldsymbol{z}_{cf} - \boldsymbol{z}_i)$. This can be converted to the change in the original dimension space $\boldsymbol{d}_{ij} = (\boldsymbol{z}_{cf} - \boldsymbol{z}_i)\boldsymbol{V}_2'$

Determine the direction and magnitude of contributing variables



The gives the direction and magnitude of the required change to reach the local decision boundary along each variable axis. A larger vector component implies a greater contribution to the prediction and change required.

Why Biplots Give Better Counterfactuals

Five advantages over existing optimisation and gradient-based methods

Visual & intuitive

The decision boundary and counterfactual are directly visible on the biplot. The impact of each variable axis on the required change is immediately apparent.

Consistent — not heuristic

The method always produces the same counterfactual for the same input. No randomness, no sensitivity to initialisation, no local minima problems.

Model-agnostic & always finds a solution

Works for any binary classifier with tabular data. Does not require gradients or model internals. Unlike analytic methods, always returns a counterfactual.

Covers the entire dataset

Counterfactuals are found simultaneously for all observations — not just one at a time. This enables both local and global interpretation in a single pass.

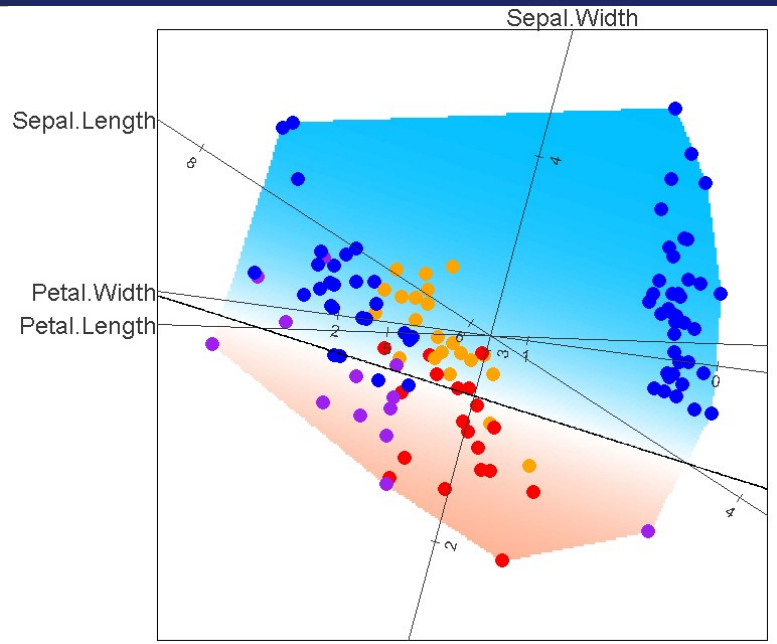
Local and global interpretations

Individual counterfactuals explain single predictions. Aggregated distances across all observations provide global feature importance — a novel contribution.

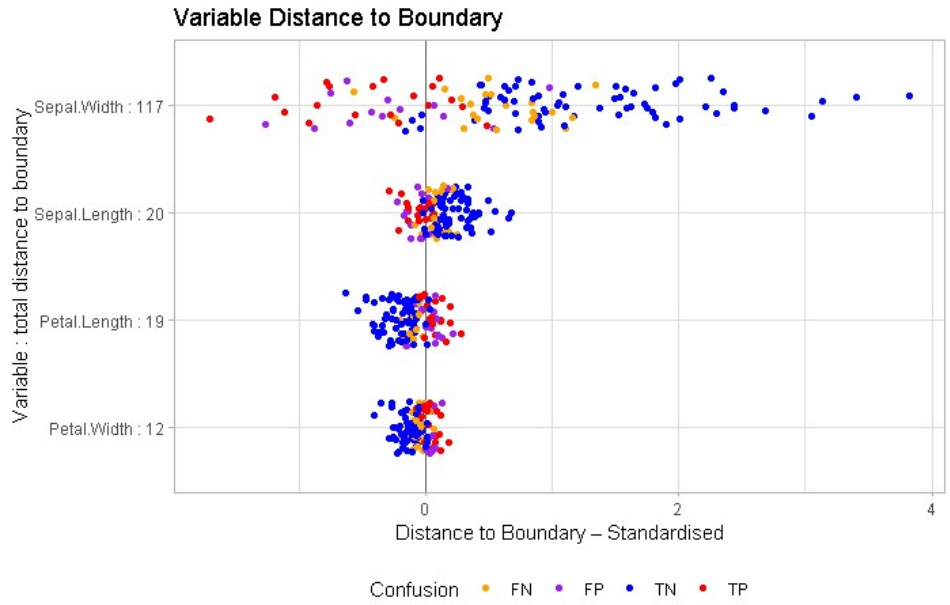
Application 1 — Global Feature Importance

Variable distance to the boundary across all observations reveals which features drive the model

PCA Biplot of the GLM model – with 70% accuracy



Variable Distance to Boundary — Jitter Plot (all observations)



Variable ranking

The number beside each variable is its total distance score — Sepal Width (117) is the most important variable; Petal Width (12) the least.

Direction

Positive distance from boundary = observation must decrease the variable value to reach its counterfactual Negative = must increase. Direction varies by class.

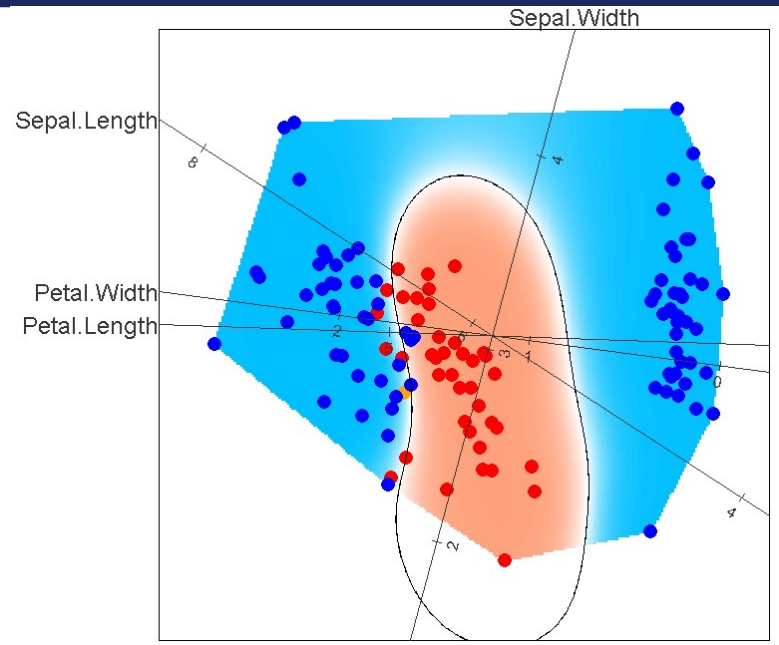
Confusion colouring

N (orange) and FP (purple) indicate the systematic pattern of misclassifications.

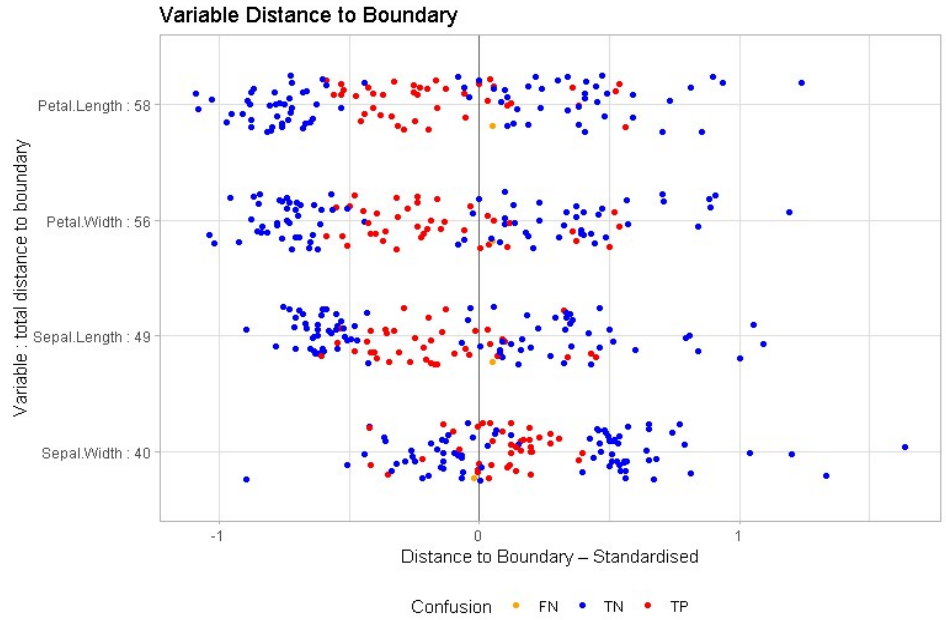
Application 1 — Global Feature Importance

Variable distance to the boundary across all observations reveals which features drive the model

PCA Biplot of the SVM model – with 95% accuracy



Variable Distance to Boundary — Jitter Plot (all observations)



Variable ranking
Change in variable importance reflects the change in die decision boundary with a SVM model – Sepal Width becomes the least important variable

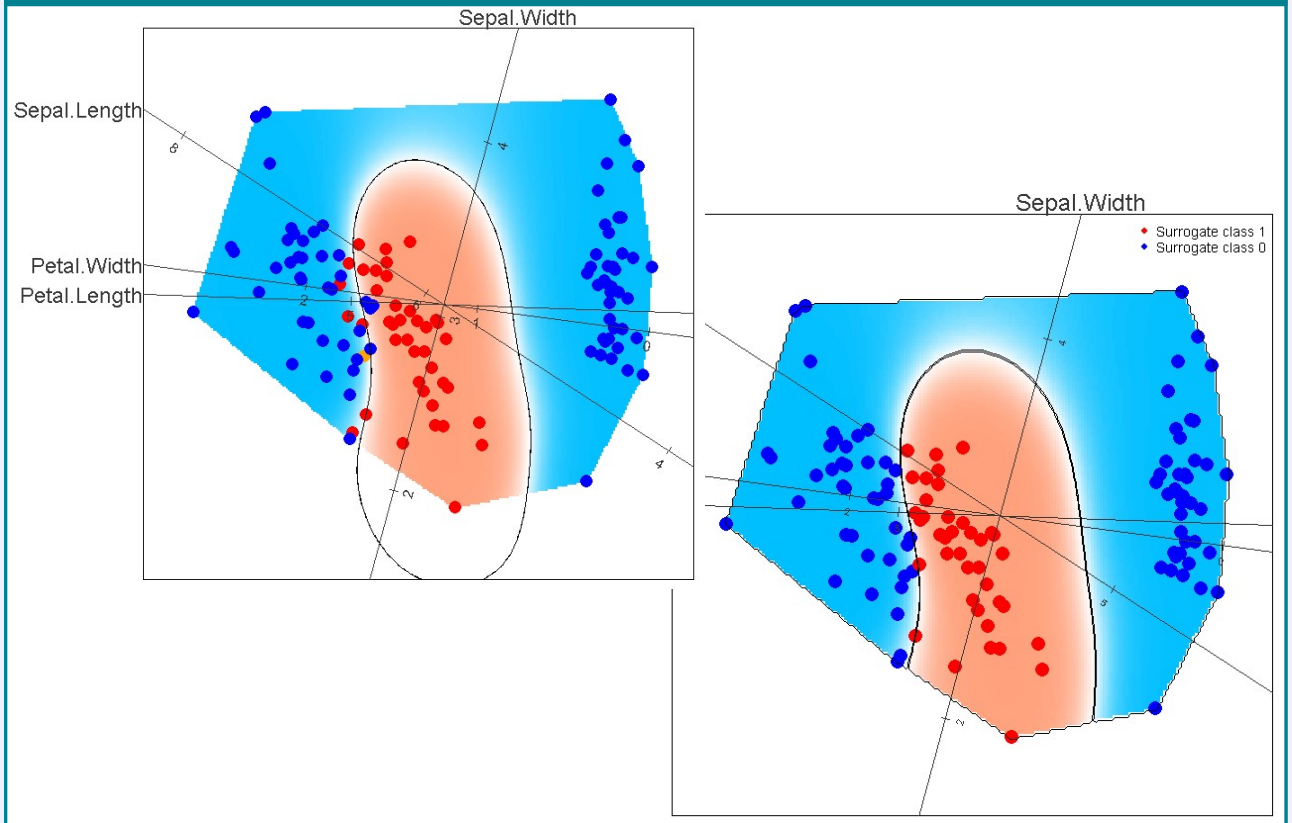
Direction
Positive (red) classified data most require variable increase to reach nearest decision boundary. Negative Classified (blue) requires both in-and-decreases

Confusion colouring
Only one data point misclassified. Data point is close to the decision boundary, where uncertainty is the greatest

Application 2 — A High-Fidelity Surrogate Model

The biplot prediction regions form an interpretable surrogate with high global fidelity

Surrogate vs Black-Box Predictions



What the Surrogate Does

What it is

Each region of the biplot (pink = class 1, blue = class 0) is a prediction zone. The surrogate assigns a class to each observation based on which zone it falls in.

Left biplot

Shows individual observations coloured by surrogate prediction. The green points are outliers — outside the training polygon, assigned NA.

Right biplot

Shows the same data with the smoothed surrogate prediction surface — a clean, continuous approximation of the black-box boundary.

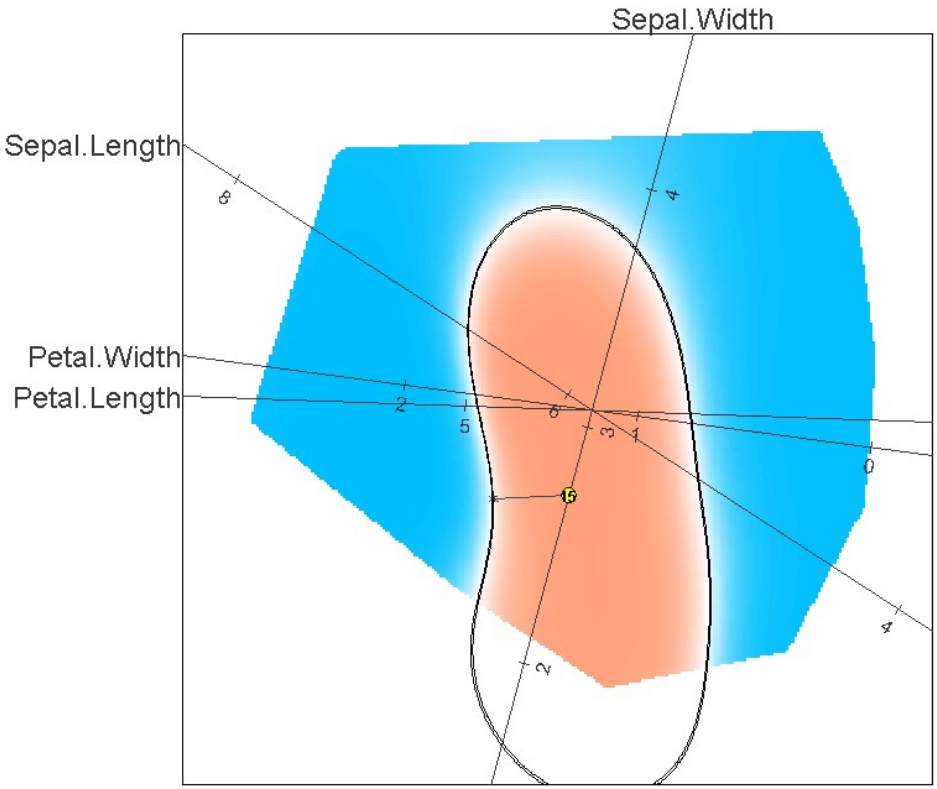
Why it matters

High fidelity surrogate = reliable global interpretation. The biplot regions explain model behaviour visually without any mathematical internals.

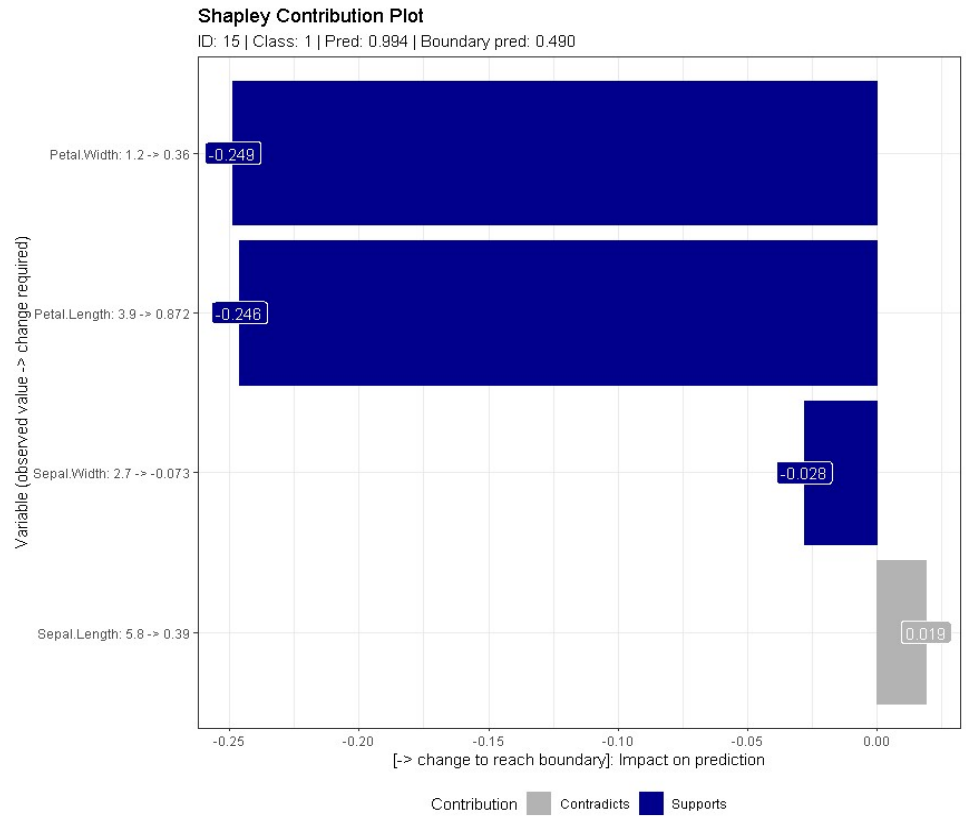
Application 3 — Local Interpretation via Shapley Values

The local biplot shows the observation, its counterfactual, and the boundary. The Shapley value decomposes the contributions

Individual Biplot and Counterfactual value for data point nr15



Shapley Contribution Plot decomposes the contribution of each variable



Limitations of Biplot Estimation of Counterfactuals

Every approximation has a cost — here is what this method trades off

2D approximation of the boundary

The counterfactual is found in a 2D projection — closer counterfactuals may exist in the full p-dimensional space. The method finds the best approximation, not the guaranteed global minimum.

Shapley computation scales with variables

Exact Shapley calculation is exponential in the number of variables. For large feature sets, permutation approximation is used — introducing some estimation error. However, because it is local (single point), computation time remains manageable.

Distance-to-boundary as importance proxy

Using boundary distance as a feature importance metric is a fast alternative to Shapley values across the full dataset — but it is not as precise. It is a useful ranking tool, not an exact attribution.

Current scope: binary, continuous, tabular

The method currently handles binary outcomes with continuous variables and tabular data. Extensions to multi-class problems, ordinal variables, and regression outcomes are directions for future work.

boundary_logic Github Account

Download the R package

Web address

Github:
https://github.com/AI-Rowan/boundary_logic

In R:
> devtools::install_github("AI-Rowan/boundary_logic", build_vignettes = TRUE)
> library(boundarylogic)
> vignette("Boundary_Logic_Pima_diabetes_workflow", package = "boundarylogic“)

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QR Code